Indian Institute of Technology, Bhilai

CSL351: Computer Networks

Assignment - 5 (NS-3 Advanced)

Full Marks: 50 *Deadline: 02-03-2026*

Submission Instructions:

1) Answer all the questions.
2) Deliverables in a .zip:
    a) Submission Guidelines: For each part, create a separate folder. Upload all the folders and Assignment Report in .zip.
    b) Readable Report **[5 Marks for report quality]** enumerating steps followed with screenshots for each of the important steps.
        i) Link to download the collected Pcap trace and mention the command/tool used.
        ii) Put the screenshots in the report for better clarity
        iii) Your report should have screenshots of your final results necessarily and you are expected to explain in the report which function was used or which line of code set the data rate or a TCP flow or a UDP flow or how did you fetch the address of a particular node, so everything needs to be carefully explained in your report.
        iv) Explain everything that has been asked in your report and highlight the important parts, and attach each and every codefile, script or anything which you used to complete the tasks.
    c) For all the experiments, write the inferences that you have observed.
3) Follow the instructions carefully given in the question.
4) You can use latex/word to make the pdf.
5) The naming of the zip file should strictly follow the given format: <Roll_No>_<Name>_<Assignment No>. If your name is Alex, your roll number is B23CS055, then the filename should be: B23CS055_Alex_04.zip
6) **Any plagiarism case will be considered unethical practice, and appropriate action will be taken against them.**

Objectives:

1. **Develop Proficiency in NS-3:**
    a. Gain hands-on experience in setting up network simulations using the ns-3 simulator.
    b. Understand how to create custom network topologies and configure various network elements.

2. **Understand Traffic Flow Configuration:**
    a. Learn to establish and analyze TCP and UDP traffic flows across different network setups.
    b. Explore the impact of link latency, data rates, and other parameters on network performance.
3. **Analyze Network Performance Metrics:**
    a. Measure key metrics such as throughput, total packets lost, and Packet Delivery Ratio (PDR) using tools like FlowMonitor and Wireshark.
    b. Visualize and interpret performance data through graphs and plots.
4. **Explore Error Models:**
    a. Implement error models like RateErrorModel to simulate packet loss and network errors.
    b. Study the impact of varying error rates and error units (Packet, Bit, Byte) on network reliability and performance.
5. **Automate Simulations:**
    a. Develop scripts to automate simulation tasks, such as data collection and processing.
    b. Generate reproducible and accurate results through iterative experiments using RngRun values.
6. **Familiarize with Visualization Tools:**
    a. Use NetAnim to visually represent and analyze network topologies and traffic flows.
    b. Capture and interpret pcap traces to understand packet-level communication.
7. **Report and Document Findings:**
    a. Document the steps followed, code implementation, and observations comprehensively.
    b. Include screenshots, flow graphs, and scripts to support results and inferences.

Getting Our Hands Dirty **[40 points]**

Using demo.cc as reference, try and understand how to set up different data rates for different links, how to print routing tables of routers and how to finally visualize the topology and simulation visually through NetAnim.

PART A: **[18 Points]**

Create a custom network topology as defined below: **[5 Points]**
The topology consists of multiple nodes connected via routers. Here's the structure:

1. **n0**, **n1**, and **n2** are connected to router0.
2. **router0** is connected to both **router1** and **router2**.
3. **n3**, **n4**, and **n5** are connected to **router1**.
4. **n6** and **n7** are connected to **router2**.

Once you have created the topology, it's time to set up the traffic flows:

1. Set up a TCP flow between **n7** and **n4**.
2. Set up a UDP flow between **n1** and **n5**.
3. Set up additional TCP flows between **n2**,**n6** and **n0**,**n3**.

** Clearly mention the **delays** and **data rates** of the links you choose and ensure that the flows between **n2** and **n6** and **n0** and **n3** have **different** values set for these than the rest of the traffic flows.

Tasks to be done:

1. Plot the flow vs throughput graph. **[3 Points]**
2. Plot the flow vs goodput (Not what you get from FlowMon()) graph. **[3 Points]**
3. Plot the flow vs total packets lost graph. **[3 Points]**
4. Generate a file 'routing-tables.txt' wherein you print the routing tables of the 3 routers. **[3 Points]**
5. Generate the NetAnim trace file and append that as well. **[1 Point]**

** **Points to be taken note of:**

- Generate the plots using gnuplot or matplotlib or any other framework and share the script.
- Append Screenshots and mark clearly the routers and nodes and the flows.
- Support your FlowMon statistics with screenshots of results wherever possible.
- The flow graphs indicate the results obtained from flowMon statistics and each flow corresponds to the ones set in the topology like between n7 and n4 or n1 and n5 so there should be 4 such flows overall, 3 of them TCP and one UDP so they come on the X-axis and ensure to record values of flows from client to sink app i.e the downlink flow as flowMon also records the reverse flows, make a note of this.

PART B: [22 Points]

Modify your source code by including an error model.
For this part you have to set up **UdpEchoClient and **UdpEchoServer** Application flows in **place of all the flows** established in PART A.

Error models are used to indicate that a packet should be errored, according to the underlying (possibly stochastic or empirical) error model. NS-3 offers the following error models (refer References section for more details).

- RateErrorModel
- ListErrorModel
- ReceiveListErrorModel
- BurstErrorModel

E.g. Rate Error Model has these parameters

- ErrorRate (0,1)
- ErrorUnit [Packet, Bit, Byte]

Run the following commands to know the available attributes of these error models and their default values.
./ns3 --run "scratch/myfirst --PrintAttributes=ns3::RateErrorModel"
./ns3 --run "scratch/myfirst --PrintAttributes=ns3::BurstErrorModel"

Though there is supposed to be an error rate for a given link (e.g., point-to-point channel), NS-3 requires you to set an error rate for each receiver (e.g., NetDevices like PointToPointNetDevice).

The following command shows the default error model for PointToPointNetDevice which is "0" meaning error-free channel.
./ns3 --run "scratch/myfirst --PrintAttributes=ns3::PointToPointNetDevice"

You need to select the error model as RateErrorModel and tune its parameters to measure Packet Delivery Ratio (PDR) in your code by transmitting at least 100 packets in each run of the simulation. As we are interested in the average behavior of the network, repeat each experiment by varying the RngRun value 10 times (i.e. 10 seed values) as one of the command line arguments. Refer to simple-error-model.cc and fifth.cc (refer to the Reference file paths which follows to know about their path in your NS-3 installation) to know how to set the error model for a NetDevice. To ensure the same error rate for both devices connecting to a point-to-point channel, set the same ErrorRate value for both in your program.

PDR is the ratio of successfully delivered packets at the receiver to the total transmitted packets by the sender. In this task, you need to measure the PDR of the network by summing up the total no. of packets received at UdpEchoServer and UdpEchoClient (say R_Total) and summing up the total no. of packets transmitted by UdpEchoClient and UdpEchoServer (say T_Total). Then PDR=R_Total/T_Total **[5 Points]**

For this part you have to set up EchoClient and EchoServer flows in place of all the flows established in **PART A.

Finally, plot variation in average PDR (on the Y-axis) by varying ErrorRate on the scale of 0 to 1 (with a step value of 0.2 on the X-axis) for Rate Error Model by using tools like Gnuplot. **[5 Points]**

Fill in below table with PDR by transmitting at least 100 packets in each run of your code:

RngRun/Error Rate	ER1=0.2	ER2=0.4	ER3=0.6	ER4=0.8	ER5=1
RngRun1=					
RngRun2=					
...					
RngRun10=					
AVG. PDR					

Tasks to be done:

- Generate such tables for each of the 3 error units [Packet, Bit, Byte], so 3 tables and corresponding plots become one of the deliverables (You have to plot avg. PDR Vs ER by using values from the above table)... **[9 Points]**
- You need to write a script that gets the data and populates the tables for the 10 seed values, and you are free to use any language for automation, bash/python, but attach the script used. **[3 points]**
- Make sure to automate all your simulation tasks and attach the scripts used.
- How to select RngRun values for the above table?
    - RngRun1 = “Last TWO DIGITS of your ROLL NUMBER”
    - RngRun2 = RngRun1+1
    - ….
    - RngRun10=RngRun1+9

** **ns-3 version should be >=3.36**

** **Reference File Paths:**

- ns-allinone-3.36.1/ns-3.36.1/examples/error-model/simple-error-model.cc
- ns-allinone-3.36.1/ns-3.36.1/examples/tutorial/fifth.cc

Check Web sources for more information.